



BIOSTAT 620

Assignment 1 - Exploratory Data Analysis

Due Date

This assignment is due by 11:59pm Eastern Time on Friday, February 13th, 2026.

Learning Goals

- Download, read, and get familiar with an external dataset.
- Step through the EDA “checklist” presented in class
- Practice making exploratory plots

Assignment Description

We will work with air pollution data from the U.S. Environmental Protection Agency (EPA). The EPA has a national monitoring network of air pollution sites that measure particulate matter (PM) concentrations. A primer on particulate matter air pollution can be found [here](#).

The primary question you will answer is whether daily concentrations of PM_{2.5} (particulate matter air pollution with aerodynamic diameter less than 2.5 μm) decreased in Michigan over the 20 years spanning from 2005 to 2025.

Your assignment should be completed in Quarto and all code should be included.

Steps

1. (20 points) Given the formulated question from the assignment description, you will now conduct EDA Checklist items 1-5. First, download 2005 and 2025 PM_{2.5} data for all sites in Michigan from the [EPA Air Quality Data website](#), then read the data into R. For each of the two datasets, check the dimensions, headers, footers, variable names and variable types. Check the distribution of the key variable we are analyzing (PM_{2.5}). Write up a summary of all of your findings.
2. (10 points) Combine the two years of data into one data frame. Use the `Date` variable to create a new column for year, which will serve as an identifier. Change the names of the key variables so that they are easier to refer to in your code.
3. (20 points) Create a basic map (or maps) in either `leaflet` or `ggplot` showing the locations of the monitoring sites, using different colors for each year. Summarize the spatial distribution of the sites. Does this distribution change from 2005 to 2025?
4. (20 points) Check for any data issues such as missing or implausible values of PM_{2.5} in the combined dataset.

Let's look more closely at the methods used for data collection. Look at the distribution of the method code variable. Examine the two most common method codes in the [EPA PM_{2.5} Codetable](#), and describe your findings. Calculate the proportion of missing values and method code for each year and report any temporal patterns you see in these observations. Why might these temporal patterns matter?

5. (30 points) Explore the main question of interest at three different levels of spatial resolution. Create data visualizations (e.g. boxplots, histograms, line plots, violin plots) **and** summary statistics that best suit each level of the analysis. Be sure to write up explanations of what you observe at each level. For levels 2 and 3, include at least one spatial plot.
 - Level 1: State. Examine the primary question for the entire state.
 - Level 2: County. Examine the primary question for every county in Michigan.
 - Level 3: City. Restrict the data to sites in Wayne county and examine the primary question for every site.

Reminder: after you upload your final rendered document to GitHub, you should download it to make sure that it looks right! If you haven't included `embed-resources: true` in the YAML header, none of your figures will show up!

Another reminder: GitHub is not (generally) intended for sharing data, so if you upload the dataset to your GitHub repo, you will lose 5 points. You can avoid this problem by storing the data somewhere else on your local machine (outside of your repo), or by adding the data file to your `.gitignore` file.

This homework has been adapted from the [case study](#) in Roger Peng's [Exploratory Data Analysis with R](#)

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LICENSE
View the [source at GitHub](#).